RCPSPTT (definition)

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1 Problem definition

The Resource-Constrained Project Scheduling Problem with Trasfer Times (RCPSPTT) is an extension of the RCPSPTT problem defined in [TODO KNUST OR QUILANT WHO FIRST], which in turn is an extension of the famous RCPSP problem defined in [TODO OLDEST REF]. In RCPSP, activities are executed using available resources, whereas the precedence relationships between them have to be adhered to, and the total makespan is minimized. Compared to the RCPSP, RCPSPTT introduces transfer times between activities. This means that resources do not have to be directly available after activity that uses them finishes, but rather a transfer of resources to the following activity might be required. The transfer time depends on both the activities and the resource transferred.

Formally, the RCPSPTT problem is defined as follows:

- Let $\mathcal{A} = \{0, 1, \ldots, \hat{\mathcal{A}}\},\ \hat{\mathcal{A}} \in \mathbb{N}$, be a set of activities with indices $i, j \in \mathcal{A}$. (RCPSP)
- Let $P \in \mathbb{N}^{|\mathcal{A}|}$ be a vector of activity processing times. (RCPSP)
- Let $\mathcal{E} = \{(i, j) \ \mid \ \text{activity } i \text{ precedes activity } j\}$ be a set of precedences. (RCPSP)
- Let $\mathcal{R} = \{0, 1, \ldots, \hat{\mathcal{R}}\},\ \hat{\mathcal{R}} \in \mathbb{N}$, be a set of primary resources indexed by $r \in \mathcal{R}$. (RCPSP)
- Let $C \in \mathbb{N}^{|\mathcal{R}|}$ be a vector of primary resource capacities. (RCPSP)
- Let $\mathbf{Q} \in \mathbb{Z}_{\geq 0}^{|\mathcal{A}| \times |\mathcal{R}|}$ be the activity-resource consumption matrix, with entries $\mathbf{Q}_{i,r}$ denoting the activity i consumption of the resource r ($i = 0 \Rightarrow \mathbf{Q}_{i,r} = C_r$). (RCPSP)
- Let $\Delta \in \mathbb{Z}_{\geq 0}^{|\mathcal{A}| \times |\mathcal{A}| \times |\mathcal{R}|}$ be a matrix of transfer delays, with entries $\Delta_{i,j,r}$ denoting the delay required if resource r is transferred between activities i and j. (RCPSPTT)

2 ILP model

The model variables are as follows:

- $a_i,\ \forall i \in \mathcal{A}$: Integer variable representing the start of the activity i, with a lower bound equal to ES_i and an upper bound equal to LF_i. (RCPSP)
- $x_{i,j},\ \forall (i,j) \in \mathcal{A} \times \mathcal{A},\ i < j$: Binary variable representing in activity i preceedes activity j.
- $f_{i,j,r},\ \forall (i,j,r) \in \mathcal{T}$: Integer variable representing the amount of resource flow r between activities i and j, with a lower bound equal to 0 and an upper bound equal to $U_{i,j,r}$. (RCPSPTT)
- $y_{i,j,r},\ \forall (i,j,r) \in \mathcal{T}$: Binary variable that represents if there exists a resource flow r between activities i and j. (RCPSPTT)

The model constraints are as follows:

$$\min\ a_{|\mathcal{A}|-1} \tag{1}$$

Precedence constraints (RCPSP):

$$\begin{aligned} a_j &\geq a_i + p_i \\ x_{i,j} &= 1 \\ x_{j,i} &= 0 \\ \forall (i,j) &\in \mathcal{E} \end{aligned} \tag{2}$$

Start time separation (optional RCPSP):

$$\begin{aligned} a_j - a_i &\geq p_i \cdot x_{i,j} - H \cdot (1 - x_{j,i}) \\ \forall (i,j) &\in \mathcal{A} \times \mathcal{A},\ i < j \end{aligned} \tag{3}$$

Flow conservation into activity (RCPSP):

$$\begin{aligned} \sum_{j:(j,i,k) \in \mathcal{T}, \Delta_{j,i,k} > 0, k = r} f_{j,i,k} &= \mathbf{Q}_{i,r} \\ \forall i \in \{1, \ldots, |\mathcal{A}| - 1\}, \forall r &\in \{0, \ldots, |\mathcal{R}| - 1\} \end{aligned} \tag{4}$$

Flow conservation out of activity (RCPSP):

$$\begin{aligned} \sum_{j:(i,j,k) \in \mathcal{T}, \Delta_{i,j,k} > 0, k = r} f_{i,j,k} &= \mathbf{Q}_{i,r} \\ \forall i \in \{0, \ldots, |\mathcal{A}| - 2\}, \forall r &\in \{0, \ldots, |\mathcal{R}| - 1\} \end{aligned} \tag{5}$$

Flow existence inference (RCPSPTT):

$$\begin{aligned} f_{i,j,r} \leq y_{i,j,r} \cdot U_{i,j,r} \\ f_{i,j,r} \geq y_{i,j,r} \\ \forall (i,j,r) \in \mathcal{T},\ i \neq |\mathcal{A}| - 1 \vee j \neq 0 \end{aligned} \tag{6}$$

Temporal linking of activities and transfers (RCPSPTT):

$$\begin{aligned} a_j - a_i \geq (p_i + \Delta_{i,j,r}) \cdot y_{i,j,r} - H \cdot (1 - y_{i,j,r}) \\ \forall i, j, r \in \mathcal{T},\ i \neq |\mathcal{A}| - 1 \vee j \neq 0 \end{aligned} \tag{7}$$

3 CP model

The model variables are as follows:

- $a_i, \ \forall i \in \mathcal{A}$: Interval variable representing activity i, with length p_i, start from interval [ES_i,LS_i] and end from interval [EF_i, LF_i]. (RCPSP)
- $f_{i,j,r}, \ \forall (i,j,r) \in \mathcal{T}, \ \Delta_{i,j,r} = 0$: Integer variable representing the amount of resource flow r between activities i and j, with a lower bound equal to 0 and an upper bound equal to $U_{i,j,r}$. (RCPSPTT)
- $z_{i,j,r}, \ \forall (i,j,r) \in \mathcal{T}$: Optional interval variable representing potential transfer of resource r between activities i and j, with length $\Delta_{i,j,r}$, start from interval [ES_i,LS_i - $\Delta_{i,j,r}$] and end from interval [EF_i + $\Delta_{i,j,r}$, LS_i]. (RCPSPTT)

The model constraints are as follows:

$$\min \ \mathrm{end}(a_{|\mathcal{A}|-1}) \tag{8}$$

Precedence constraints (RCPSP):

$$\textsc{EndBeforeStart}(a_i, a_j) \qquad \forall (i,j) \in \mathcal{E} \tag{9}$$

Resource flow for the source node $i = 0$ (RCPSPTT):

$$\sum_{j:(0,j,r)\in\mathcal{T}} f_{0,j,r} = C_r \tag{10}$$

Implicating transfer activation (only if $\Delta_{i,j,r} = 0$) (RCPSPTT):

$$f_{i,j,r} \geq 1 \ \Rightarrow \ \textsc{PresenceOf}(z_{i,j,r}) = 1 \qquad \forall (i,j,r) \in \mathcal{T}, \Delta_{i,j,r} = 0 \tag{11}$$

Flow conservation into activity (RCPSPTT):

$$\sum_{j,k:(j,i,k)\in\mathcal{T},\Delta_{j,i,k}>0,k=r} \textsc{HeightAtStart}(z_{j,i,k}, \textsc{Pulse}(z_{j,i,k}, (0, U_{j,i,k}))) + \sum_{j,k:(j,i,k)\in\mathcal{T},\Delta_{j,i,k}=0,k=r} f_{j,i,k} = \mathbf{Q}_{i,r} \quad \forall i \in \{1,\ldots,|\mathcal{A}|-1\}, \forall r \in \{0,\ldots,|\mathcal{R}|-1\} \tag{12}$$

Flow conservation out of activity (RCPSPTT):

$$\sum_{j,k:(i,j,k)\in\mathcal{T},\Delta_{i,j,k}>0,k=r} \textsc{HeightAtStart}(z_{i,j,r}, \textsc{Pulse}(z_{i,j,r}, (0, U_{i,j,r})))$$
$$+ \sum_{j,k:(i,j,k)\in\mathcal{T},\Delta_{i,j,k}=0,k=r} f_{i,j,k} = \mathbf{Q}_{i,r}$$
$$\forall i \in \{0,\ldots,|\mathcal{A}|-2\}, \forall r \in \{0,\ldots,|\mathcal{R}|-1\} \quad (13)$$

Temporal linking of activities and transfers (RCPSPTT):

$$\begin{aligned} &\textsc{EndBeforeStart}(a_i, z_{i,j,r}) && \forall (i,j,r) \in \mathcal{T} \\ &\textsc{EndBeforeStart}(z_{i,j,r}, a_j) && \forall (i,j,r) \in \mathcal{T} \end{aligned} \quad (14)$$

Resource capacity during activities (cumulative constraint) (RCPSPTT):

$$\sum_{i,k:\mathbf{Q},\mathbf{Q}_{i,k}>0,k=r} \textsc{Pulse}(a_i, \mathbf{Q}_{i,r}) + \sum_{(i,j,k)\in\mathcal{T},\Delta_{i,j,k}>0,k=r} \textsc{Pulse}(z_{i,j,r}, (0, U_{i,j,r})) \leq C_r$$
$$\forall r \in \{0,\ldots,|\mathcal{R}|-1\} \quad (15)$$